

# LeaseOS Canada Hours-of-Service Compliance Logbook v2

Provincial Alberta + federal interprovincial + over-160-km RODS + Canada-U.S. cross-border design controls

**Version:** 2026-09-17. **Purpose:** LeaseOS / Claude project knowledge and implementation specification.

**Scope note:** This version fully specifies the Alberta/federal decision path and adds a U.S. cross-border transition section. Other Canadian provinces and territories are represented as separate jurisdiction profiles to be populated/verified from their current legislation before production use.

**Core principle:** The 160 km rule is a record-keeping exemption, not permission for extra hours. If a driver does not qualify for the applicable 160 km exemption, the driver must maintain the required record of duty status (RODS/daily log), and federally regulated drivers who are required to keep RODS generally require a certified ELD unless another regulatory exception applies.

## 1. LeaseOS operating-mode decision tree

| Question                                                                                                 | Yes                                                                                             | No                                                                                                                              |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Is the carrier/operation federally regulated (interprovincial/territorial or international undertaking)? | Use federal Canadian HOS/ELD profile while operating under Canadian rules.                      | Use the applicable provincial/territorial intra-provincial profile.                                                             |
| Does the driver meet every applicable 160 km exemption condition?                                        | Normal daily RODS/ELD may be exempt; carrier time records are still mandatory.                  | RODS/daily log required. For federally regulated drivers, certified ELD is generally required unless another exception applies. |
| Does the trip enter the United States?                                                                   | Activate U.S. cross-border profile at the border and validate U.S. HOS/ELD eligibility/history. | Remain on Canadian jurisdiction profile.                                                                                        |

## 2. Alberta intra-provincial 160 km exemption

For Alberta's provincial daily-log exemption, all applicable conditions must remain true: the driver does not operate beyond a **160 km radius from the home terminal**; returns to the home terminal and is released from work within **15 hours** from the start of the work shift; and the carrier retains accurate start/release time records for **6 months**.

When any condition stops applying, LeaseOS must change the day to **LOG\_REQUIRED**. Alberta's carrier-training material says the driver must commence a daily log and record the total on-duty hours accumulated during the immediately preceding 7 days.

## 3. Federal Canadian 160 km exemption

Federal section 81 exempts the driver from the normal RODS requirement only when the driver drives, or is instructed to drive, within a **160 km radius of the home terminal**; returns to the home terminal each day to begin at least **8 consecutive hours off duty**; the carrier keeps accurate/legible cycle and on-duty records plus supporting documents for at least **6 months**; and the driver is not operating under a permit (unless the regulation/exemption provides otherwise).

**Important distinction:** The federal rule is tied to the home terminal and its statutory conditions. LeaseOS should not replace this with a generic '160 km from current location' test.

## 4. When the driver goes beyond 160 km

Crossing the applicable radius, or otherwise losing an exemption condition, does **not** create extra driving time. It changes the record-keeping state. LeaseOS should immediately require a full RODS/daily log for that day and reconstruct the earlier duty status/history needed by the applicable rule.

| LeaseOS event                                                 | Required action                                                                                                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Route/geofence predicts travel beyond 160 km before departure | Start day as LOG_REQUIRED/ELD_REQUIRED where applicable; do not wait for the boundary crossing.                                                                                                |
| Unexpected dispatch takes driver beyond 160 km mid-shift      | Convert to LOG_REQUIRED; prompt driver to complete/manual-enter prior duty information required for continuity.                                                                                |
| Federal driver was radius-exempt but now must keep RODS       | Require certified ELD unless another federal ELD exception applies. Transport Canada says a driver who did not keep RODS at the start of the day may need to manually enter prior HOS history. |
| Missing location/history needed to prove exemption            | Fail closed: do not claim RADIUS_EXEMPT. Require the more complete record.                                                                                                                     |

## 5. Federal Canadian HOS controls - south of 60

| Control          | LeaseOS enforcement                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 13 hours driving | No driving after accumulating 13 hours driving unless the required qualifying off-duty period/rest calculation permits driving again. |
| 14 hours on duty | No driving after accumulating 14 hours on-duty time in the applicable work-shift calculation.                                         |
| 16 elapsed hours | No driving after the 16th elapsed hour between qualifying core off-duty periods, subject to sleeper-berth calculations.               |
| Daily off-duty   | Track the statutory daily off-duty requirement and any valid deferral/sleeper exception separately.                                   |
| Cycle 1          | 70 on-duty hours in 7 days; track reset eligibility.                                                                                  |
| Cycle 2          | 120 on-duty hours in 14 days plus the additional statutory 7-day control; track reset eligibility.                                    |

## 6. Sleeper berth

For a single driver under federal Canadian rules, qualifying sleeper time may be accumulated in no more than two periods when neither is shorter than 2 hours and the two total at least 10 hours. The paired periods must also satisfy the driving, on-duty and elapsed-time tests in section 18. Sleeper berth changes the calculation; it is not a blanket extension of legal hours.

**LeaseOS:** store both qualifying sleeper periods, pair ID, start/end timestamps, total duration, preceding/following driving, preceding/following on-duty time, elapsed-time calculation, and VALID/PENDING/INVALID result.

## 7. Federal ELD requirements and exceptions

Transport Canada states that federally regulated commercial drivers who must complete RODS must use a compliant, third-party-certified ELD. Regulatory exceptions include specified permit/exemption operations, qualifying rentals of 30 days or less, pre-2000 model-year vehicles, and drivers who are not required to keep RODS.

The vehicle should carry the required ELD information packet, including user instructions, data-transfer instructions, malfunction instructions and enough blank RODS for at least 15 days. Carrier RODS and supporting documents are retained for at least 6 months.

**LeaseOS product boundary:** LeaseOS may manage HOS workflows and integrate with an ELD, but it must not present itself as the legally required Canadian ELD unless the actual hardware/software implementation is tested and certified under the Transport Canada framework.

## 8. Full RODS / daily log - required when exemption does not apply

This is the LeaseOS data model for a full log day. Production ELD output must meet the applicable Canadian technical standard and regulation.

|                          |  |                             |                |
|--------------------------|--|-----------------------------|----------------|
| Driver                   |  | Date                        |                |
| Co-driver                |  | Start of day                |                |
| Carrier                  |  | Cycle                       | 1 / 2          |
| Home terminal            |  | Principal place of business |                |
| Unit(s)                  |  | Trailer(s)                  |                |
| Start odometer           |  | End odometer                |                |
| Distance driven          |  | Personal-use distance       |                |
| ELD identifier           |  | Exemption/exception         | None / specify |
| Shipping docs / manifest |  | Time zone                   |                |

|                       | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | Total |
|-----------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|
| Off duty              |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |
| Sleeper berth         |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |
| Driving               |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |
| On duty - not driving |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |       |

**Duty-status events / annotations** - every status change should preserve time and location plus required remarks/supporting data.

| Time | Location | Duty status / event | Annotation / supporting document |
|------|----------|---------------------|----------------------------------|
|      |          |                     |                                  |
|      |          |                     |                                  |
|      |          |                     |                                  |

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Driver certification/signature: \_\_\_\_\_

## 9. Radius-exempt carrier time record

Use this instead of a normal daily RODS only when every applicable radius-exemption condition is satisfied.

|                               |                         |                           |                |
|-------------------------------|-------------------------|---------------------------|----------------|
| Driver name                   |                         | Carrier                   |                |
| Date                          |                         | Home terminal             |                |
| Unit #                        |                         | Trailer #                 |                |
| Rule set                      | AB provincial / Federal | Cycle                     | 1 / 2 / N/A    |
| Shift start                   |                         | Shift end / released      |                |
| Total on-duty                 |                         | Total driving             |                |
| Maximum radius                |                         | Returned to terminal?     | Yes / No       |
| 8 consecutive hours off next? | Yes / No / N/A          | AB release within 15 h?   | Yes / No / N/A |
| Permit/exemption involved?    | No / specify            | Supporting docs retained? | Yes / No       |

Driver/carrier certification: \_\_\_\_\_ Date: \_\_\_\_\_

## 10. Canada interprovincial / provincial-border operation

A carrier that is an extra-provincial undertaking is governed by the federal Canadian HOS regime for covered operations. LeaseOS should therefore determine the carrier's regulatory status, not simply switch to each province's intra-provincial HOS rule whenever GPS crosses a provincial border.

**Routing behavior:** Before dispatch, classify the trip as INTRAPROVINCIAL, INTERPROVINCIAL/TERRITORIAL or INTERNATIONAL. Preserve the classification in the job/trip audit record. A border-crossing route should trigger federal HOS/ELD validation before the driver accepts dispatch.

## 11. Canada-U.S. international operation

Canadian drivers entering the United States must comply with applicable U.S. FMCSA HOS/ELD requirements while operating in the U.S. FMCSA guidance states that a Canada/Mexico-domiciled driver operating in the U.S. who must use RODS for more than 8 days in any 30-day period is subject to the U.S. ELD rule unless another exception applies.

FMCSA also recognizes the U.S. **150 air-mile short-haul exception** for qualifying cross-border operations when all U.S. short-haul conditions are met. Current FMCSA cross-border guidance describes return to the normal work reporting location and release within **14 hours** as part of that exception.

**Do not reuse Canada's 160 km test for the U.S.** The U.S. short-haul radius is 150 air miles and has its own statutory conditions. LeaseOS should treat the border as a jurisdiction transition and evaluate U.S. rules independently.

| At/Before Canada-U.S. border           | LeaseOS check                                                                                                                      |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Driver's current Canadian RODS/history | Preserve/export the history needed to establish prior duty status and make it available for inspection.                            |
| U.S. ELD applicability                 | Determine whether U.S. ELD rule applies, including the more-than-8-days-in-30 rule and any valid exception.                        |
| U.S. short-haul claim                  | Validate 150 air-mile radius, normal work reporting location, return/release requirement and all other U.S. short-haul conditions. |
| Return to Canada                       | Re-evaluate Canadian federal HOS/ELD state and ensure cross-border duty events remain continuous/auditable.                        |

## 12. Other Canadian provinces and territories - production architecture

The following profiles must exist separately in LeaseOS. This v2 document does **not** assert that each jurisdiction uses Alberta's 160 km/15-hour provincial exemption. Each profile must be verified against current provincial/territorial law before enabling enforcement decisions.

| Jurisdiction profile                                          | Required fields before production                                                                                                                              |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| British Columbia                                              | Intra-provincial applicability, HOS limits, daily-log/radius exemption, ELD adoption/enforcement, retention, permits.                                          |
| Saskatchewan                                                  | Same jurisdiction-specific controls; include oilfield/special permit handling where applicable.                                                                |
| Manitoba                                                      | Same jurisdiction-specific controls.                                                                                                                           |
| Ontario                                                       | Same jurisdiction-specific controls.                                                                                                                           |
| Quebec                                                        | Same jurisdiction-specific controls, including French-language regulatory/document requirements where applicable.                                              |
| New Brunswick / Nova Scotia / PEI / Newfoundland and Labrador | Separate profiles; do not collapse Atlantic provinces into one rule set.                                                                                       |
| Yukon / Northwest Territories                                 | Separate profiles and north-of-60/federal interactions where applicable.                                                                                       |
| Nunavut                                                       | Separate profile. Transport Canada notes the federal Regulations/ELD requirement are not presently applied there in the same manner; verify before production. |

## 13. LeaseOS compliance-engine states

**JURISDICTION\_UNRESOLVED, PROVINCIAL\_HOS, FEDERAL\_CANADA\_HOS, US\_HOS, RADIUS\_EXEMPT, RADIUS\_AT\_RISK, LOG\_REQUIRED, ELD\_REQUIRED, ELD\_EXCEPTION, PAPER\_MALFUNCTION\_LOG, SLEEPER\_SPLIT\_PENDING, SLEEPER\_SPLIT\_VALID, SLEEPER\_SPLIT\_INVALID, HOS\_WARNING, DRIVING\_PROHIBITED.**

Recommended fail-closed sequence: resolve carrier regulatory status -> resolve trip jurisdiction -> resolve vehicle applicability -> resolve radius exemption -> resolve ELD requirement/exception -> validate current/prior duty history -> validate sleeper/deferral/cycle -> authorize or block driving.

## 14. Official sources / verification basis

Government of Canada - Commercial Vehicle Drivers Hours of Service Regulations (SOR/2005-313), including sections 13, 18 and 81.

Transport Canada - Electronic logging devices for commercial drivers and motor carriers; ELD handout for motor carriers and drivers.

Government of Alberta / Alberta Carrier Training - Module 3, Hours of Service, including Alberta and federal logbook-exemption guidance.

U.S. Federal Motor Carrier Safety Administration - Cross Border Transportation FAQs, including U.S. ELD and 150 air-mile short-haul guidance for Canada/Mexico-domiciled carriers.

**Production warning for Claude/LeaseOS:** Do not treat this PDF as an ELD certification. Do not infer that crossing 160 km permits additional hours. Do not apply Alberta's provincial exemption nationwide. Do not apply Canada's 160 km radius rule in place of the U.S. 150 air-mile short-haul rule. Missing jurisdiction or duty-history data must fail closed.